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NO HARDWARE

v1 was rejected for hardware in the budget, an unverified cost line, an offtake contract, a 30-day RWA-token scope, and a single-operator verification risk. Each is addressed, not reworded.

2 • PROBLEM

3 • WHAT THIS SPRINT BUILDS

```

any meter —adapter—▶ interval.schema.json —▶ gates (versioned spec) —▶ batch + hash
    ▲ Modbus RTU/TCP · HTTP/JSON · CSV replay                                |
    |                                                                           ▼
    └─ anyone re-runs the same gates on the published batch ◀— Stellar Testnet
                                2-of-3 multisig: proposer + independent verifier + third signer

```

Built on shipped, open-source work: the V1-V7 validator suite with 24 adversarial tests, the verification model, the public gate demo at qwatt.org/verificacion, and the live testnet asset.

4 · DELIVERABLES

D1 · Meter-agnostic ingestion layer

25 H

- Modbus RTU/TCP adapter where the **register map is configuration, not code** — a new meter is supported by adding a JSON file, never by editing the engine.
- HTTP/JSON push adapter and CSV/file replay adapter. All normalize to a versioned **interval.schema.json**.
- Ships with three example register maps as configs: PZEM-017, EPEver Tracer series, a generic single-phase AC meter.

Acceptance: the same input, delivered through all three adapters, produces byte-identical verification output.

D2 · Single-source gate specification + conformance suite

20 H

- **gates.spec.json**: every threshold and rule (nameplate ceiling, solar-elevation gate, meter agreement, plausibility bounds) in one versioned file.
- Generated bindings for the engine and for the public browser demo — the demo runs generated code, not a hand-written copy.
- Conformance suite of `{input, verdict, reason}` fixtures executed against every binding in CI; divergence fails the build. Spec version written into every anchored batch.

Acceptance: 100% conformance across bindings; the existing 24 adversarial tests pass unchanged against the generated engine.

D3 · Multi-party authorization on Stellar

25 H

- Anchoring account with native multisig, medium threshold **2-of-3**: **proposer** (the engine) · **independent verifier** (separate service and key; downloads the *published* batch, recomputes the hash, re-runs the gates, signs only on agreement, never receives data from the proposer directly) · **third signer** held off-team, designated at kickoff.
- One atomic transaction per batch: `manageData` evidence hash + `manageData` cumulative total + a payment op against the existing testnet asset as anchoring vehicle only.

Acceptance: with the proposer alone, nothing anchors. With a batch altered after the verifier's read, nothing anchors. Both demonstrated with fixtures and linked transactions.

D4 · Public evidence feed + browser reconciliation

15 H

- Every batch — **accepted and rejected** — published as canonical JSON with hash and rejection reason.
- Reconciliation page recomputes in the visitor's browser from Horizon: anchored hashes ↔ published batches, cumulative totals, sequence gaps.

Acceptance: a reviewer with no access to the team verifies any anchored batch end-to-end from public artifacts alone.

D5 · Real-data validation without hardware

15 H

- Replay ≥ 30 days of a public PV generation dataset through D1→D4 on testnet.
- Replay the induced-fault fixtures — shaded sensor, disconnected meter, night injection, implausible ramp — every one must be rejected and logged. Public write-up linking every transaction.

Acceptance: non-empty rejection log with reasons; every accepted interval traceable to an anchored transaction; solar-elevation gate pinned in tests to Santiago's real values.

5 · BUDGET — USD 5,000

DELIVERABLE	HOURS
D1 · Ingestion layer	25
D2 · Gate spec + conformance	20
D3 · Multisig + independent verifier	25
D4 · Evidence feed + reconciliation	15
D5 · Validation + write-up	15
Total	100 h × USD 50

No hardware, hosting beyond free tiers, marketing, legal, audit, or prior work. Testnet fees are zero. Hours are those funded; the team contributes additional time at its own cost.

7 · OUT OF SCOPE

Hardware of any kind · token issuance, redemption or any RWA claim
· Soroban contracts · commercial agreements · mainnet · marketing.

8 · NEXT STEPS — THE GRANT PATH

STAGE	SCOPE	FUNDING
This sprint	Verification engine, multisig anchoring, public evidence	Instaward 1
Instaward 2	Issuance semantics on testnet: $\min(A,B)$ tokenization, floor-never-round, cumulative accounting, double-mint protection, burn on redemption	Instaward
Instaward 3	Soroban verification contract: the independent verifier becomes an on-chain contract rather than a service; QWATT-C retire semantics	Instaward
SCF Build	Physical pilot (dual-metered rooftop), assurance-level audit, legal structure, mainnet readiness	SCF

Each stage is a separate submission with its own acceptance criteria; none depends on approval of the next.

9 · MILESTONES

DAY	CHECKPOINT
7	D1 adapters + schema; three register-map configs merged
14	D2 spec, generated bindings, conformance green in CI
21	D3 multisig live on testnet; first 2-of-3 anchored batch; negative tests linked
28	D4 feed + reconciliation page live; D5 replay complete
30	Write-up published; all links in the final report

6 · SECURITY POSTURE

- **No single key can write.** Enforced by Stellar multisig, not by policy.
- **No hidden logic.** Gates are open-source, versioned, reproducible; the version is on-chain.
- **Rejections are public.** A verifier that shows only successes has demonstrated nothing.
- **Adversarial coverage.** One test per attack in the trust model; drill fixtures become permanent regressions.
- **Honest limits.** The team operates the proposer and verifier services. This design does not claim decentralization; it claims every record is independently verifiable and that authorization requires a party outside the team. Removing the team from operations entirely is the on-chain verifier of the follow-on grant.